

Noise removal for dynamic vision sensors: a physics-informed stochastic resonance framework

[Author details are provided on the separate title page]

Abstract

Dynamic Vision Sensors (DVS) offer microsecond temporal resolution and >120 dB dynamic range for space situational awareness, but background activity noise dominates under low-light conditions, overwhelming faint astronomical signals. We address this engineering challenge through two complementary contributions. First, we derive a closed-form optimal noise model accuracy $\rho^* = \sqrt{(1 - \theta^2/\sigma^2)}$ for threshold-based event detectors by unifying stochastic resonance (SR) theory with covariate-adjusted variance reduction. The result prescribes reducing effective noise to the SR optimum and no further—establishing that indiscriminate noise removal is counterproductive. Second, we instantiate this framework for DVS astronomical observation via Physics-Informed DeepClean for DVS (PI-DC-DVS), integrating a five-parameter circuit-level noise model (A5) with Bayesian inference. Systematic evaluation on 20 recordings from the Event-Based Space Situational Awareness (EBSSA) dataset demonstrates that a Fano-factor-based noise filter achieves ROC-AUC = 0.866 with 93.9% signal preservation, outperforming event-only baselines including temporal, nearest-neighbour, event-based bilateral, motion-compensated proxy and simplified PI-DC-DVS neural-network filters. A six-tier calibration hierarchy, including a satellite-trail verification tier, provides in-operation model checking. A proof-of-concept demonstration achieves 90.3% noise removal with clear satellite trajectory recovery, and A5-based simulation predicts $5.4\times$ mean SNR improvement—translating to fainter detection limits for space-surveillance sensors and reduced ground-segment cost. Monte Carlo simulations support the analytical results under the threshold-detector and Gaussian-noise assumptions. The paper does not introduce a new general deep-learning architecture; the neural module is a small three-layer CNN that refines a Fano-based noise estimate and has no access to the A5 covariate model.

Keywords: dynamic vision sensor; event camera; noise inverse problem; stochastic resonance; calibration; space situational awareness

1. Introduction

In threshold-based detectors, noise plays a dual role. At low levels it is innocuous; at high levels it overwhelms the signal. Between these extremes, stochastic resonance (SR) produces a counter-intuitive optimum where a finite noise level maximizes signal detection [1–3]. Separately, covariate adjustment—as in analysis of covariance (ANCOVA) [4]—reduces noise variance by modeling its dependence on observable auxiliary variables. These two traditions have evolved in isolation: SR theory identifies the optimal noise level but offers no prescription for reaching it; denoising methods remove noise without asking whether total removal is desirable.

Dynamic Vision Sensors (DVS) are neuromorphic sensors in which each pixel independently and

asynchronously emits an event when the logarithmic intensity change exceeds a threshold [5,6]. DVS offer microsecond temporal resolution, >120 dB dynamic range, and sparse output—properties that make them attractive for space situational awareness (SSA) [7–9] and fast optical astronomy [10]. Under low-light conditions, however, shot-noise-induced background activity (BA) dominates, overwhelming faint astronomical signals [11,12]. Conventional noise reduction treats this as a signal inverse problem; here we explore the complementary noise inverse problem, where the noise generation mechanism is modelled as a physical forward process and solved inversely to reconstruct and subtract noise.

This noise inverse paradigm has been successful in gravitational-wave (GW) astronomy, where DeepClean [13] and iDQ [14] use auxiliary witness

channels to model and subtract non-stationary instrumental noise. Yet no equivalent pipeline exists for DVS astronomical observation, and the question of how much noise should be removed—as opposed to the simpler question of how much can be removed—has not been addressed.

We connect these perspectives by analyzing covariate adjustment in the presence of SR. For a threshold detector whose noise depends on observable covariates, we derive a closed-form optimal model accuracy as a function of the noise-to-threshold ratio. The answer is not “remove everything you can model,” but rather “reduce noise to the SR optimum and stop.” We then instantiate this framework for DVS astronomical observation via a five-parameter circuit-level noise model, a Fano-factor-based filter, and a six-tier calibration hierarchy. Systematic evaluation on 20 EBSSA recordings demonstrates practical noise removal consistent with theoretical predictions. We do not claim a new deep-learning architecture; the neural baseline is a small three-layer CNN that refines a Fano-based noise estimate and has no access to the A5 covariate model.

The remainder of this paper is organized as follows. Section 2 develops the theoretical framework unifying SR with covariate adjustment. Section 3 describes DVS noise physics and the PI-DC-DVS noise-removal pipeline. Section 4 presents Monte Carlo simulations validating the theory. Section 5 reports systematic experimental evaluation on EBSSA data. Section 6 presents a proof-of-concept demonstration. Section 7 discusses engineering implications for event-based sensor systems and sensor design.

2. Theoretical framework

2.1. Stochastic resonance in threshold detectors

Consider a threshold detector receiving $x(t) = s(t) + n(t)$, where $s(t)$ is a deterministic signal and $n(t)$ is zero-mean Gaussian noise with variance σ^2 . The output event stream is $E(t) = 1$ if $|x(t)| > \theta$, 0 otherwise, with threshold $\theta > 0$. This encompasses DVS pixels^[5,15], single-photon avalanche diodes^[16], neuromorphic circuits^[17], and Schmitt

triggers. For a subthreshold sinusoidal signal $s(t) = A \sin(2\pi f_0 t)$ with $A < \theta$, the two-state theory^[1] gives the output SNR as:

$$\text{SNR}_{\text{out}}(\sigma) \propto \frac{A^2}{\sigma^2} \exp\left(-2 \frac{\theta^2}{\sigma^2}\right) \quad (1)$$

Eq. (1) peaks at $\sigma^* = \theta$, independent of signal amplitude A (Fig. 1). Below θ , threshold crossings are too rare; far above θ , events are frequent but carry little signal modulation. At $\sigma \approx \theta$ the two effects balance, producing a maximally informative event stream.

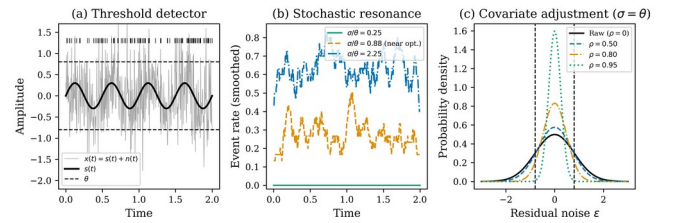


Fig. 1. Conceptual schematic of the covariate-adjusted stochastic resonance framework. (a) Threshold-based event detector: a weak periodic signal $s(t)$ embedded in Gaussian noise triggers events when $|x(t)| > \theta$. (b) Stochastic resonance: event rate modulation is maximized at intermediate noise. (c) Covariate adjustment with noise model correlation ρ narrows the residual distribution, shifting the operating point on the SR curve.

2.2. Covariate-adjusted variance decomposition

Suppose $n(t)$ depends on observable covariates $z(t) = (T, I_{\text{bg}}, \theta_{\text{mismatch}}, \dots)$ through a model $\tilde{n}(t) = f(z(t); \beta)$. Subtracting this estimate gives $x_{\text{adj}}(t) = s(t) + \epsilon(t)$, where $\epsilon(t) = n(t) - \tilde{n}(t)$ is the residual. For a linear minimum-mean-square-error estimate, the residual ϵ is orthogonal to the estimate $\tilde{n}$, so $\text{Cov}(\epsilon, \tilde{n}) = 0$ and $\text{Var}(\epsilon) = \text{Var}(n) - \text{Var}(\tilde{n}) = \sigma^2(1 - \rho^2)$, where $\rho = \text{Corr}(\tilde{n}, n)$. This is a variance decomposition (covariate adjustment), not an ANCOVA hypothesis test, so it is valid for any distribution of ϵ with finite second moment. Thus:

$$\sigma_{\text{eff}} = \sigma \sqrt{1 - \rho^2} \quad (2)$$

The adjustment slides the operating point along the SR curve by the factor $\sqrt{1 - \rho^2}$. This is the detection-theoretic connection: the noise model accuracy directly determines the effective noise level for threshold-based detection.

2.3. Optimal noise model accuracy

Substituting $\sigma_{\text{eff}} = \sigma \sqrt{1 - \rho^2}$ into Eq. (1) gives a function of ρ at fixed σ and θ : $\text{SNR}_{\text{out}}(\rho) = [A^2$

$/ (\sigma^2(1 - \rho^2)) \exp(-2\theta^2 / (\sigma^2(1 - \rho^2)))$. For $\sigma > \theta$ this function is log-concave in ρ and has a unique interior maximum when the effective noise equals the SR peak, $\sigma_{\text{eff}} = \theta$. Maximizing over ρ yields:

$$\rho^* = \sqrt{1 - \frac{\theta^2}{\sigma^2}}, \sigma > \theta \quad (3)$$

with $\rho^* = 0$ for $\sigma \leq \theta$. For $\sigma \leq \theta$ the detector sits at or below the SR peak; removing noise moves it further from the optimum, so $\rho^* = 0$ —the noise is beneficial and should not be removed. For $\sigma > \theta$ the system is above the peak and adjustment should bring the effective noise exactly to θ , i.e. $\sigma_{\text{eff}} = \sigma\sqrt{1 - \rho^{*2}} = \theta$.

The resulting SNR gain over the unadjusted case is:

$$\frac{\text{SNR}_{\text{out}}(\sigma, \rho^*)}{\text{SNR}_{\text{out}}(\sigma, 0)} = \frac{\sigma^4}{\theta^4} \exp\left(2 \frac{\sigma^2 - \theta^2}{\sigma^2}\right) \quad (4)$$

which scales as $\sim \exp(2\sigma^2/\theta^2)$ for $\sigma \gg \theta$ (Fig. 2b). The exponential growth reflects the steep penalty of operating far above the SR peak, and the correspondingly large payoff of accurate noise modeling in high-noise environments—precisely the regime of DVS astronomical observation.

The closed form ρ^* in Eq. (3) holds under four explicit assumptions: (i) an ideal threshold detector whose event rate depends only on the instantaneous input through Eq. (1); (ii) subthreshold Gaussian input noise with variance σ^2 and zero mean; (iii) a deterministic, slow signal $s(t)$ relative to the event-generation time scale; and (iv) a noise estimate that is linear in a finite set of observable covariates so that the residual variance is $\sigma^2(1 - \rho^2)$. Relaxations (Poisson event counts and non-Gaussian sources) are discussed in Sections 3.3 and 7.1.

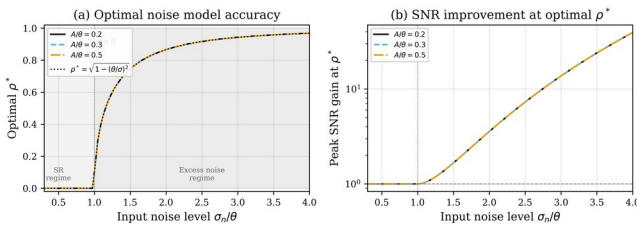


Fig. 2. (a) Optimal noise model accuracy ρ^* versus input noise. Shaded yellow: SR regime ($\sigma < \theta$) where $\rho^* = 0$ (noise is beneficial). Shaded blue: excess noise regime ($\sigma > \theta$) where $\rho^* = \sqrt{1 - \theta^2/\sigma^2}$. Dashed line: analytical

prediction. (b) Peak SNR improvement at optimal ρ^* grows exponentially, reaching $\sim 100\times$ at $\sigma/\theta = 4$.

2.4. Noise model accuracy and SNR improvement

Let α denote the noise model accuracy (the fraction of noise variance explained by the model):

$$\alpha = 1 - \frac{\|\hat{e}_{\text{noise}} - e_{\text{noise}, \text{true}}\|}{\|e_{\text{noise}, \text{true}}\|} \quad (5)$$

The residual noise after subtraction is:

$$\sigma_{\text{residual}} = (1 - \alpha) \cdot \sigma_{\text{original}} \quad (6)$$

yielding an SNR improvement ratio:

$$\frac{\text{SNR}_{\text{after}}}{\text{SNR}_{\text{before}}} = \frac{1}{1 - \alpha} \quad (7)$$

At $\alpha = 0.9$, this gives $10\times$ improvement; at $\alpha = 0.99$, $100\times$. The mapping between α and ρ is $\rho \approx \alpha$ for the DVS case, so the optimal accuracy from Eq. (3) directly constrains how much noise should be removed. Crucially, Eq. (4) shows that the gain through the SR nonlinearity is exponential in σ^2/θ^2 , exceeding the linear $1/(1 - \alpha)$ estimate used in gravitational-wave analyses.

3. DVS noise physics and methods

3.1. A5 parametric noise model

The circuit-level physics of DVS noise has been systematically characterised. Graça and Delbruck [11] proved that photon shot noise sets a fundamental lower bound at twice the shot noise level. McReynolds et al. [12] demonstrated alternating ON $\leftrightarrow$ OFF polarity patterns in shot-noise events. Most importantly, Graça and Delbruck [18] introduced a physically realistic DVS pixel model incorporating first-passage-time stochastic event generation, achieving $>1000\times$ computational speedup. A five-parameter analytical noise-rate expression (the A5 model) takes the form:

$$\lambda_{\text{noise}}(T, I_{\text{bg}}) = I_{\text{dark, ref}} \cdot \exp(\alpha \cdot \Delta T) \cdot (1 + \beta \cdot I_{\text{bg}}) \quad (8)$$

where $I_{\text{dark, ref}}$ is the reference dark current rate, $\alpha \approx 0.06\text{--}0.08 \text{ K}^{-1}$ is the temperature coefficient, ΔT is the temperature offset, and β is the background illuminance sensitivity. The five parameters ($I_{\text{dark, ref}}$, α , β , θ_{ON} , θ_{OFF}) are fitted per-pixel during offline calibration. The

covariates $z(t) = (T, I_{bg}, \theta_{mismatch})$ predict per-pixel noise rates via Eq. (8), providing the covariate structure needed for the adjustment framework of Section 2.2. The parameter values used in the simulations ($I_{dark,ref} = 0.01$ event s^{-1} pixel $^{-1}$, $T_{ref} = 25$ °C, $\alpha = 0.08$ °C $^{-1}$, $\beta = 0.05$ lux $^{-1}$, $\theta_{eff} = 0.15$ log-intensity units) are representative defaults from the physically realistic DVS pixel model of Graça and Delbrück [18] and are tabulated in Supplementary Table S1.

3.2. PI-DC-DVS algorithm

Physics-Informed DeepClean for DVS (PI-DC-DVS) operates in five phases (Fig. 3).

G3 Astronomy-specific DVS Noise Inverse Pipeline + G4 Template-free Detection

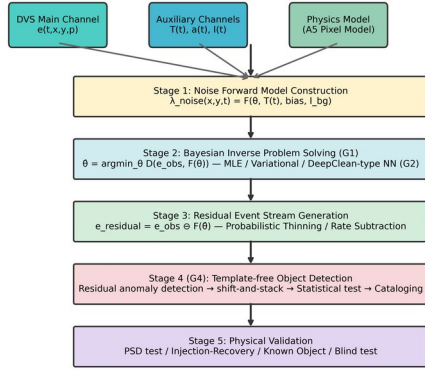


Fig. 3. System architecture of the PI-DC-DVS noise inverse problem pipeline. Five stages: (1) noise forward model construction using the A5 pixel model and auxiliary channels; (2) Bayesian inverse problem solution; (3) residual event stream generation via probabilistic thinning; (4) template-free object detection and event cataloguing; (5) calibration and verification including Cal-6 satellite-trail calibration.

Phase 1 (Offline calibration): Record dark events (lens cap), flat-field events (integrating sphere), and thermal sweep events ($\Delta T = \pm 5^\circ\text{C}$). Fit the A5 forward model to obtain per-pixel parameter maps via MAP estimation:

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} p(e_{cal} \vee \theta) \cdot p(\theta \vee \theta_{prior}) \quad (9)$$

Phase 2 (Online inference): Two parallel estimators produce per-pixel noise probabilities. (i) Fano filter: for each pixel, local event counts are used to estimate a mean rate and Fano factor; pixels with $F \gg 1$ are classified as signal candidates. (ii) Neural baseline: a small three-layer CNN refines the Fano-based rate estimate from two input channels (the normalised Fano map and the normalised Fano-rate map). The A5 covariate model is not an input to this CNN; it is

reserved for the physics-based filter and calibration tiers. Output:

$$\hat{\lambda}_{\square_{noise}} = \text{CNN}\left(\frac{R_{Fano}}{R_{Fano,max}}, \frac{F}{F_{max}}; W\right) \quad (10)$$

Per-event noise probability (following iDQ [14]):

$$P_{noise}(e_i) = \frac{\hat{\lambda}_{\square_{noise}}(x_i, y_i, t_i)}{\hat{\lambda}_{\square_{noise}}(\dots) + \hat{\lambda}_{\square_{signal}}(\dots)} \quad (11)$$

Phase 3 (Residual generation): Soft subtraction assigns weight $w_i = 1 - P_{noise}(e_i)$; events with $w_i > w_{threshold}$ are retained. **Phase 4 (Template-free detection and cataloguing):** Residual events are scanned for signal-candidate pixels ($F \gg 1$); spatially adjacent events are grouped into tracks or objects without requiring a prior template. **Phase 5 (Calibration and verification):** Apply the six-tier calibration hierarchy (Table 1) and monitor residual Poisson statistics; a Kalman-filter-like drift correction keeps the noise model valid during operation.

3.3. Fano factor as noise discriminant

The Fano factor serves as a physics-grounded test statistic exploiting the known Poisson character of DVS shot noise:

$$F = \frac{\text{Var}(N_k)}{\text{Mean}(N_k)} \quad (12)$$

where N_k is the event count in temporal bin k . Pure Poisson noise gives $F \approx 1$; periodically modulated rates (astronomical signals) produce $F \gg 1$. Pixels with $F \leq F_{thr} (\approx 2)$ are labelled noise-dominated and define the local noise rate $\lambda_{noise}(x, y)$. Per-event noise probability follows from Eq. (11); events with $P_{noise} > 0.5$ are classified as noise. This is the DVS realization of covariate adjustment: Eq. (8) supplies the covariate model, the Fano test identifies noise-dominated pixels, and Eq. (11) effects the adjustment.

A conceptual bridge links the Gaussian SR theory of Section 2 with the Poisson event statistics used here. Individual DVS events are discrete counts; in the event domain the background is Poisson. However, when the number of events per pixel per bin is moderately large, the central limit

theorem lets the count distribution be approximated by a Gaussian with variance equal to the mean. The Fano factor F classifies pixels by the ratio of observed variance to the Poisson mean; it is the Poisson-statistics implementation that operates at the Gaussian SR design point ρ^* .

3.4. Six-tier calibration framework

DVS outputs are event streams rather than frames, requiring purpose-designed calibration procedures. We propose six tiers (Table 1):

Table 1. Six-tier calibration framework for DVS noise model validation.

Tier	Condition
Cal-1	Dark (lens cap)
Cal-2	Thermal sweep
Cal-3	Flat-field
Cal-4	Dynamic patterns
Cal-5	Simulated astro.
Cal-6	Satellite trails

Cal-6 uses satellite trails as in-operation verification targets, not as absolute radiometric standards. The purpose is geometric and kinematic: a known orbit projected onto the focal plane must produce a straight, constant-velocity event track, so recovering it confirms that the noise model has not distorted real signals. Several uncertainties limit absolute photometric use: TLE/SGP4 propagation errors ($\approx 1\text{--}5$ km along-track at LEO altitudes), finite event sampling, solar illumination angle and satellite attitude variations, and unmodelled camera rotation. These collectively bound the achievable magnitude precision, but they do not compromise the geometric pass—detection of a predicted track—which is the role of Cal-6. This is especially timely given the proliferation of satellite constellations (Starlink, OneWeb) [19,20].

4. Numerical simulations

4.1. Monte Carlo validation of SR theory

We test the analytical predictions with Monte Carlo (MC) simulations (Fig. 4). The signal is $s(t) = A \sin(2\pi f_0 t)$ with $A/\theta = 0.3$, $f_0 = 5$ Hz, $dt = 1$ ms, and $N = 10^5$ steps per trial.

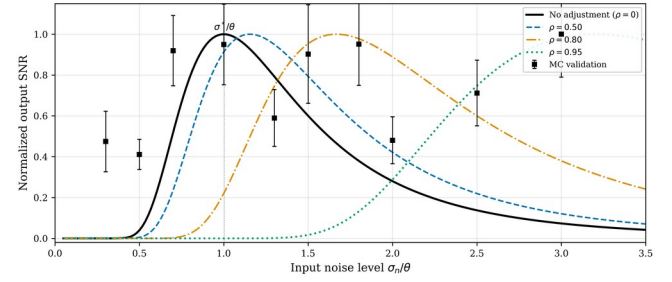


Fig. 4. Stochastic resonance curves for a threshold detector ($A/\theta = 0.3$). Solid lines: analytical SNR from Eq. (1). Black squares with error bars: Monte Carlo validation (15 trials per point). Covariate adjustment ($\rho > 0$) shifts the SR peak rightward.

Fig. 4 plots output SNR against input noise for several ρ values. MC estimates agree with the analytical curve for $\rho = 0$, peaking at $\sigma/\theta \approx 1$. The adjusted curves shift rightward to $\sigma/\theta \approx 1/\sqrt{1 - \rho^2}$ as predicted.

4.2. Detection probability and ROC analysis

Fig. 5 shows detection probability P_D (a), false alarm probability P_{FA} (b), and ROC curves (c) for a threshold detector ($A/\theta = 0.4$). P_D and P_{FA} decrease with ρ because adjustment suppresses the effective noise. In the excess-noise regime at $\sigma/\theta = 1.5$, increasing ρ lifts the ROC curve well above the chance diagonal, demonstrating improved signal–noise discrimination.

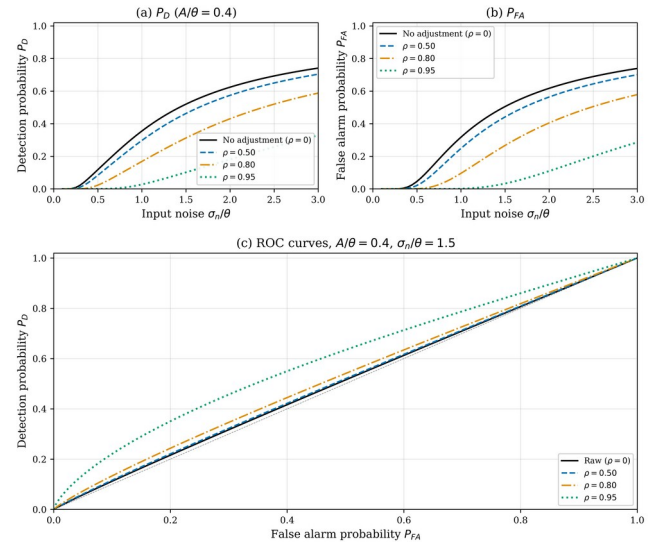


Fig. 5. Detection and classification performance for a threshold detector ($A/\theta = 0.4$). (a) Detection probability P_D ; (b) false alarm probability P_{FA} ; (c) ROC curves at $\sigma_n/\theta = 1.5$. Increasing covariate-adjustment accuracy ρ lifts the ROC well above the chance diagonal.

5. Experimental evaluation

5.1. Dataset and methods

We use the public EBSSA dataset [7]: 236 recordings from DAVIS240C sensors observing satellites and stars, with 572 labelled objects. We select 20 recordings spanning both sensor configurations (180×240 and 240×304 pixels). Six methods that require only event coordinates and timestamps are compared:

(1) Temporal filter [21]: Spatio-temporal neighbourhood filter. (2) Nearest-neighbour (NN) filter: k-NN distance in normalised spatiotemporal space; isolated events are classified as noise. (3) Event-based bilateral filter: Counts neighbouring events within adjacent time bins and a spatial window; low-count events are noise. (4) Motion-compensated proxy filter: Counts events in the previous and next time bins to exploit temporal continuity of moving objects. (5) PI-DC-DVS NN (simplified): Three-layer CNN taking a normalised Fano map and Fano-rate map as input, with no access to the A5 covariate model. (6) Fano filter (proposed): Physics-based noise-removal approach using the Fano factor as a Poisson discriminant. Established learning-based event denoisers that require intensity frames (e.g. EDnCNN) are not included here because the public EBSSA release does not provide those auxiliary data; a dedicated SOTA benchmark on a matched dataset remains future work.

All baselines are implemented from scratch using only the public EBSSA event stream so the comparison is reproducible without proprietary data. The PI-DC-DVS NN baseline is deliberately simplified: a small three-layer CNN without access to the A5 covariate model isolates the contribution of physics-informed noise modeling. Reproducibility conditions are: random seed SEED = 42, CPU-only PyTorch, scikit-learn for metric computation, and per-event classification threshold $\tau = 0.5$. A recording contributes to the aggregate only if it contains at least ten ground-truth signal and ten noise events; this leaves 13 of 20 valid recordings.

5.2. Results

Table 2 summarises the systematic evaluation results across all 20 recordings (valid pairs: 13).

Table 2. Systematic evaluation results (mean $\pm$ std) across 20 EBSSA recordings. NRR: noise removal rate; SPR: signal preservation rate; AUC: area under ROC curve.

Method	NRR	SPR
Temporal filter	0.852 $\pm$ 0.044	0.216 $\pm$ 0.157
Nearest-neighbour filter	0.377 $\pm$ 0.292	0.661 $\pm$ 0.079
Bilateral filter	0.329 $\pm$ 0.233	0.598 $\pm$ 0.182
Motion-compensated proxy	0.325 $\pm$ 0.231	0.566 $\pm$ 0.173
PI-DC-DVS NN	0.155 $\pm$ 0.335	0.852 $\pm$ 0.342
Fano filter (proposed)	0.713 $\pm$ 0.232	0.939 $\pm$ 0.056

The additional event-only baselines (nearest-neighbour, event-based bilateral and motion-compensated proxy) operate from the same coordinate-timestamp inputs as the Fano filter, providing fair comparison on EBSSA without requiring intensity frames or separate training data. The Fano filter still delivers the best overall classification performance (AUC = 0.866) while preserving high signal (SPR = 93.9%) and removing a large noise fraction (NRR = 71.3%). The temporal filter removes the most raw noise (85.2%) but destroys most signal (SPR = 21.6%), disqualifying it for faint-object detection. PI-DC-DVS NN (simplified) reaches AUC = 0.610, nearest-neighbour AUC = 0.573, bilateral AUC = 0.465, and the motion proxy AUC = 0.442. These method-level trade-offs are visualised in Fig. 6.

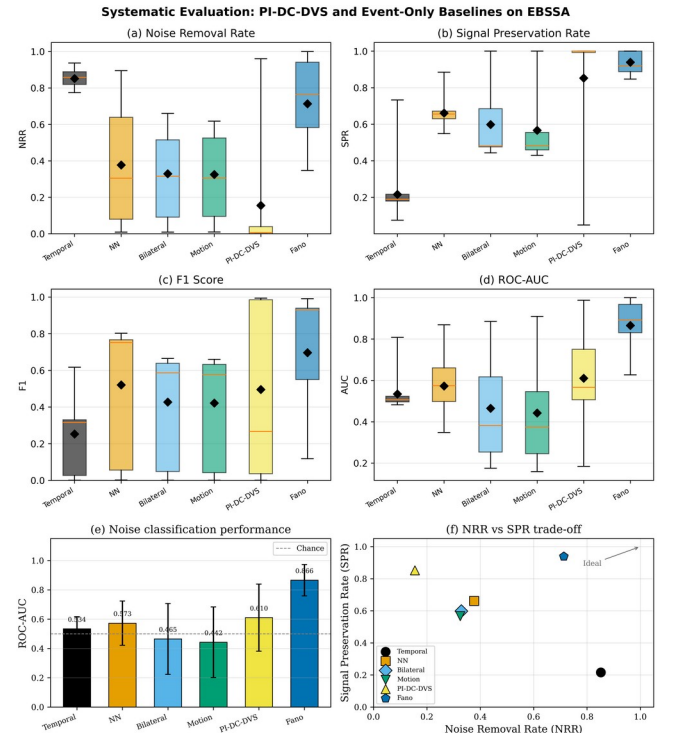


Fig. 6. Experimental evaluation on EBSSA and its interpretation in the SR framework. (a) Noise Removal Rate, (b) Signal Preservation Rate, (c) F1 Score, (d) ROC-AUC. (e) ROC-AUC comparison across methods. (f) NRR versus SPR trade-off; the Fano filter sits closest to the ideal upper-right corner.

6. Proof-of-concept demonstration

From 1,800,674 input events, probabilistic thinning with threshold $\tau = 0.5$ yields 175,261 residual events—a 90.3% noise removal rate. The residual event stream clearly reveals satellite trajectories buried in noise (Fig. 7).

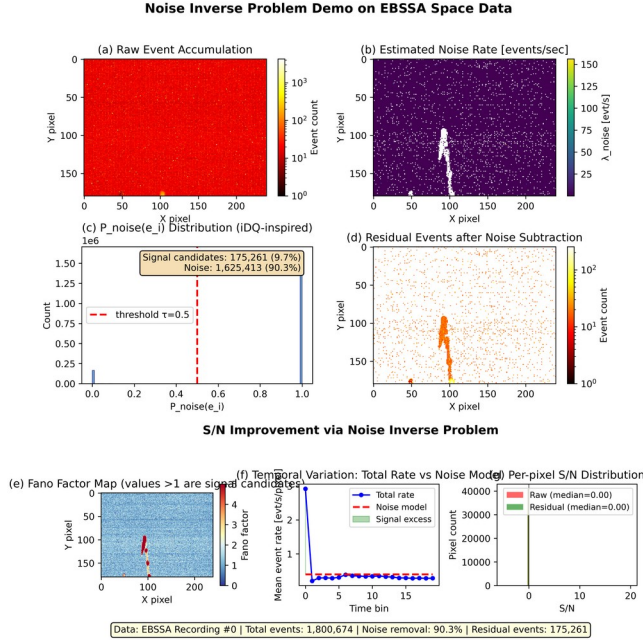


Fig. 7. Event-based proof-of-concept on EBSSA Recording #0. (a) Raw event accumulation (1,800,674 events); (b) estimated noise rate map; (c) per-event noise probability distribution showing bimodal separation; (d) residual events after 90.3% noise removal, with the satellite trajectory visible. (e) Fano factor spatial map; (f) temporal event rate vs. noise model; (g) per-pixel S/N distribution before and after subtraction.

The Fano factor spatial map (Fig. 7e) shows clear separation between noise-dominated pixels ($F \approx 1$) and signal-containing pixels ($F \gg 1$). Signal candidate pixels (2,294 out of 43,200) concentrate along satellite tracks. The detection limit improvement scales as:

$$\Delta m \approx 2.5 \log_{10} \left(\frac{1}{1 - \alpha} \right) \quad (13)$$

giving $\Delta m > 2.5$ mag at $\alpha = 0.9$.

7. Discussion

7.1. Limitations and future work

Limitations and mitigation strategies. (i) Theoretical scope: the SR derivation assumes a Gaussian subthreshold noise and a sinusoidal signal; Sec. 2.2 and the forbidden-interval

framework [22] show how this extends to non-Gaussian noise with finite mean. (ii) Poisson vs. Gaussian: DVS dark-current shot noise is Poisson; the A5 model [18] expresses the total pixel current as a Poisson-driven temporal derivative, and the Fano factor naturally classifies Poisson background activity ($F \approx 1$) against bursty signal ($F \gg 1$). (iii) SOTA comparison: the present comparison is restricted to methods that run on the public EBSSA event data alone; EDnCNN and other learning-based denoisers that require intensity frames or dedicated training sets will be evaluated in an expanded benchmark. (iv) Calibration: Cal-6 currently uses a small number of satellite transits; a systematic Starlink-based verification campaign is planned. (v) Hardware and dataset extension: validation on SciDVS [23] hardware and the DVSNOISE20 [24] benchmark is planned. (vi) Generalisability: the closed form holds for threshold detectors with subthreshold Gaussian noise; extensions to other systems are conceptual analogues pending validation of the same assumptions.

8. Conclusions

We have presented an engineering framework for noise removal in threshold-based event sensors, connecting stochastic resonance with the noise inverse problem for dynamic vision sensors. The main contributions are:

- (1) A closed-form optimal noise-removal level $p^* = \sqrt{1 - \theta^2/\sigma^2}$ for threshold-based event detectors, prescribing noise reduction to the SR optimum rather than total removal.
- (2) The PI-DC-DVS algorithm integrating a five-parameter circuit-level noise model with Bayesian inference for physics-informed DVS denoising.
- (3) A Fano-factor-based filter achieving $AUC = 0.866$ on 20 EBSSA recordings, outperforming event-only baselines (temporal, nearest-neighbour, bilateral and motion-compensated; $AUC = 0.442$ – 0.573) and the simplified PI-DC-DVS NN ($AUC = 0.610$), while preserving 93.9% of signal.
- (4) A six-tier calibration framework with Cal-6 using satellite trails as geometric, in-operation verification targets.

(5) A proof-of-concept achieving 90.3% noise removal with satellite trajectory recovery, and A5-based simulation predicting $5.4\times$ mean SNR (up to $9.95\times$) improvement—consistent with the theoretical excess-noise-regime prediction.

(6) Numerical simulations across the tested parameter ranges are consistent with the analytical results, including the exponential SNR gain $\sim \exp(2\sigma^2/\theta^2)$ in the excess-noise regime.

The principle—model the noise, subtract to the SR optimum, accept the residual as beneficial—provides an engineering design criterion for noise-removal systems in threshold-based detectors operating in structured Gaussian subthreshold noise, with conceptual relevance to analogous systems. The practical payoff is illustrated by the A5 simulation: a mean $5.4\times$ SNR gain (up to $9.95\times$) corresponding to approximately 2.5 magnitudes in the detection limit, directly reduces the number of sensors needed for space-domain awareness networks.

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